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Efficient tick-shape networks of full-residual point-depth-point blocks for image classification

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Abstract

Light-weight convolutional neural networks (CNNs) are crucial for deploying computer vision applications in mobile devices thanks to their compact models, small computational complexity, and energy efficiency. However, such models ordinarily have steady-increased channels in their backbone leading to a sharp increase in the model size; while the deficiency of identity mappings in their residual mechanism can lead to modest performance in feature extraction. To mitigate those issues, we propose light-weight networks based on three novel concepts as follows. Firstly, an efficient perceptron is presented to encapsulate point-depth-point (PDP) features extracted by light-weight convolutions along with a full-residual (FR) mechanism through the architecture of a network. This full-residual connection is proposed to deal with the shortcoming of the existing light-weight models whose architecture has incompletely exploited identity mappings. It is due to the variability of spatial dimension caused by several strides of their convolutional operations. Secondly, a tick-shape backbone is then introduced by designing its structure in accordance with the channel elasticity of the FR-PDP perceptron subject to the shape of a check mark. Thirdly, taking advantage of the channel elasticity concept, three tick-shape networks (TickNets) are constructed in a light-weight architecture by hooking one or more tick-shape backbones. Experimental results for image classification on benchmark datasets have clearly corroborated the efficacy of the proposed methods. The code implementation of tick-shape models is available at <https://github.com/nttbrk25/TickNets>.

Keywords: Light-weight models, depthwise/pointwise convolutions, CNN-based networks, Full-residual mechanism, Image classification.

1. Introduction

CNN-based networks with powerful learning abilities have been introduced to handle different tasks in computer vision such as semantic segmentation [1, 2, 3, 4, 5], image classification [6, 7, 8, 9], object detection [10, 11, 12], real-life issues [13, 14, 15, 16, 17], etc. [In order to reach](#)

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high performance, the architecture of the CNN-based networks is usually designed as a very deep structure with multiple convolutional layers. Such designing way raises a large volume of learnable weights as well as expensive computational costs. It can be one of the serious barriers when addressing such expensive models for real applications in mobile devices and automotive systems with computing resources and energy storage restricted in practice. Addressing this problem, many efforts have been made to construct various light-weight architectures with a slight volume of parameters in lower complexity. In general, the light-weight architectures are structured in a sequential order of perceptron blocks. Therein, several CNN-based operations with cheap computation are taken into account for the learning process, e.g., depthwise and pointwise convolutions. These cheap convolutions are often arranged into different incorporations and then encapsulated to form a unified perceptron block for feature extraction. Indeed, MobileNets [18, 19, 20] are some of the most popular networks with shallow layers by taking advantage of light-weight operations (i.e., depthwise and pointwise convolutions) instead of the standard convolutional operations. ShuffleNets [21, 22] were proposed to address several grouped pointwise convolutions along with shuffled channels for depthwise convolutions. Tan and Le [23] structured versions of EfficientNets by taking advantage of the linear bottleneck blocks (LiBs) of MobileNetV2 [19] in different sizes of convolutional kernels, while MnasNets [24] are formed by using factorized hierarchical search space with light-weight convolutions. It can be observed that the backbone of these small networks is designed subject to the steady increase of #channels. Going with such deep channels in the last layers can sharply enlarge the volume of learnable parameters, as done in the backbone of MobileNetV1. To mitigate this issue, the LiB-based features [19] are exploited in several small models [20, 24, 23] with steady-increased backbones in slight #channels. Furthermore, motivated by the residual mechanism [25], the insertion of a shortcut connection is taken into account for several perceptron blocks of the aforementioned networks to mitigate the problems of vanishing/exploding gradients. However, there is a deficiency of identity mappings in their residual connection (so-called semi-residual) due to the different dimension of feature maps during the learning progresses. This difference is caused by the change of strides in several convolutional operations utilized to solidify the spatial features of a given tensor for further learning processes or by the change of #channels between two corresponding layers.

Addressing the above problems, we introduce tick-shape networks of full-residual point-depth-point blocks based on main operations as follows. For an input tensor $\mathcal{T}$, an efficient perceptron is proposed to take advantage of light-weight convolutions for the point-depth-point (PDP) features of $\mathcal{T}$ (see Fig. 1 for a quick view). To deal with the shortcoming of the non-identity mappings caused by the dimension change of feature maps, we propose a full-residual (FR) connection by taking advantage of the flexible property of a pointwise convolution to handle the input tensor $\mathcal{T}$ for the corresponding stretchiness of the spatial information and channels. It means that the pointwise operation will transform $\mathcal{T}$ so that the size of the resultant volume is identical to that of the PDP output for a full-residual mechanism (named FR-PDP). A tick-shape backbone is then proposed by locating the channel elasticity of FR-PDP blocks to be agreed with the shape of a check mark. Taking advantage of the channel elasticity concept, three tick-shape networks (TickNets) are investigated for performance improvement by hooking one or more tick-shape backbones, correspondingly. To the best of our knowledge, it is the first time that a light-weight CNN-based model can be designed subject to the fluctuation of #channels of an output perceptron, instead

of the steady-increased #channels in the conventional models [18, 19, 20]. Experimental results for image recognition on benchmark datasets have validated that TickNets achieve good results in comparison with other light-weight models. In short, the major contributions of this work can be described as

- An efficient FR-PDP block is proposed to exploit point-depth-point features with a full-residual mechanism.
- A tick-shape backbone is introduced subject to the channel elasticity of FR-PDPs.
- Three light-weight TickNets of FR-PDPs are investigated for enhancing the learning ability in image representation.
- Experimental results for image classification on benchmark datasets have validated the good performance of TickNets compared to others.

2. Related works

2.1. Depthwise and pointwise convolutions

For a given tensor $\mathcal{T} \in \mathbb{R}^{C_{in} \times H \times W}$ with C_{in} channels and $H \times W$ dimension for each channel, a conventional convolution will convert $\mathcal{T}$ into another tensor as

$$\mathcal{T}_\mu = \mu_i * \mathcal{T}, \quad i = 1, \dots, C_{out} \quad (1)$$

where “*” means a convolutional operator; μ_i denotes a $C_{in} \times \mathcal{K} \times \mathcal{K}$ kernel; C_{out} means the number of output channels expected from the operation. By applying such C_{out} kernels, the resultant tensor will be formed as $\mathcal{T}_\mu \in \mathbb{R}^{C_{out} \times H \times W}$. Usually, the learnable weights of those kernels can be arbitrarily valued in the initial stage, and they would be then optimized via the mechanism of backpropagation in each training step. From now on, we would like to disregard the spatial dimension of channels in the concerning tensor for a simple presentation.

In order to reduce the number of learnable parameters, light-weight convolutions (e.g., depthwise and pointwise) are often exploited in several small CNN-based architectures [18, 23, 21]. Indeed, a depthwise convolution [26] is separably defined for each spatial channel c_j of $\mathcal{T}$ as

$$\mathcal{T}_d = \xi_j * c_j, \quad j = 1, \dots, C_{in} \quad (2)$$

where ξ_j denotes a $1 \times \mathcal{K} \times \mathcal{K}$ kernel. As a result, the obtained tensor $\mathcal{T}_d$ has the same dimension as the input one, i.e., $\mathcal{T}_d \in \mathbb{R}^{C_{in} \times H \times W}$. It can be seen that $\mathcal{T}_d$ is merely an output of a filtering operation. In other words, new features have not been produced by the depthwise convolution at its learning progress. So a pointwise convolution is often addressed after the depthwise execution. It is defined as a convolutional operator with $C_{in} \times 1 \times 1$ kernels ζ_j applied to the points in $\mathcal{T}_d$ as

$$\mathcal{T}_p = \zeta_j * \mathcal{T}_d, \quad j = 1, \dots, C_{out} \quad (3)$$

The obtained tensor $\mathcal{T}_p$ would be in a size of $C_{out} \times H \times W$, i.e., $\mathcal{T}_p \in \mathbb{R}^{C_{out} \times H \times W}$. Furthermore, as the definition of the pointwise convolution, it is absolutely utilized for expansion or contraction of the channel cardinality for an expected output tensor by addressing various values of C_{out} , correspondingly.

2.2. Light-weight convolutions for CNN-based networks

Taking into account depthwise/pointwise convolutions, many light-weight networks have been introduced to be in accordance with the computational limitation of mobile devices. Indeed, thanks to addressing the light-weight convolutions, the complexity of a DSC-based operation in MobileNetV1 [18] is about nine times lower than that of the standard convolution, i.e., correspondingly convoluting with a 3×3 kernel. Meanwhile, the LiB-based blocks of MobileNetV2/V3 [19, 20] can flexibly produce compressed features by addressing pointwise convolutional operations. Motivated by the efficacy of MobileNets, several efforts have been made to structure other small networks for further enhancement. Indeed, Xception [27] is structured by substituting the standard convolutions of Inception modules [28] for depthwise separable convolutions, correspondingly. Zhang *et al.* [21] proposed ShuffleNet by taking into account the shuffled features of a pointwise-based tensor which has been computed by a grouped pointwise convolution. Ma *et al.* [22] then moved the shuffle operation for the final features of a ShuffleNet-based unit to form ShuffleNetV2. In the meanwhile, Tan *et al.* [24] took into account a reinforcement-learning-based search algorithm for optimizing blocks of light-weight convolutions to form MnasNet, while EfficientNet [23] can obtain good performance owing to systematically studying model scaling and identifying to balance network depth, width, and resolution. Recently, Haase and Amthor [29] presented blueprint separable convolutions (BSConvs) based on intra-kernel correlations for a more efficient separation of the light-weight convolutional operations. MobileNeXt [30] is structured by sequencing multiple sandglass blocks in which two depthwise convolutions have been utilized inside for encoding more expressive spatial information. Different from the previous approaches, we propose in this work an efficient light-weight convolutional perceptron with a full-residual connection (FR-PDP) so that it can produce non-expanded features to maintain the depth channel cognition of an input tensor. TickNets are then designed by addressing FR-PDPs subject to tick-shape backbones of output channels.

Furthermore, several efforts tried to modify some layers of the existing small networks to boost the performance with a reasonable increase of computational complexity for various real applications: depthwise dilated convolutions [31, 32, 33], atrous pointwise convolutions [34], contextual pointwise operation [35], depthwise-based features with attention mechanism [20, 23, 36, 37]. In another aspect, several methods have been proposed to reduce the computational complexity by modifying the standard convolutions of a large network. Wang *et al.* [38] presented factorized CNNs to reduce the dense connection between the input and output channels by using a topological subdivision technique. Lu *et al.* [39] introduced a repeated group convolutional kernel to decrease #parameters by considering the redundancy of group convolutions in multiple scales.

2.3. Semi-residual mechanism for light-weight networks

In practice, the depthwise and pointwise convolutions are usually exploited in different ways of sequential orders and encapsulated to form perceptron blocks of light-weight networks for feature extraction. For instance, a sequential structure of one depthwise and one pointwise forms the DSC block in the architecture of MobileNetV1 [18], while the feature expansions of the light-weight convolutions are taken into account to form the LiB block in MobileNetV2 [19]. To capture more spatial information, two depthwise convolutions are addressed in the sandglass block of MobileNext [30]. Furthermore, motivated by the residual mechanism [25], the insertion of a

shortcut connection is taken into account for several perceptron blocks [19, 20, 21, 22, 23, 30] to mitigate the problems of vanishing/exploding gradients. In these light-weight networks, the residual mechanism will connect an input tensor $\mathcal{T}_{in}$ with an output one $\mathcal{T}_{out}$ in the case of the same dimension (so-called a semi-residual operation), i.e., $\mathcal{T}_{semi_res} = \mathcal{T}_{in} \oplus \mathcal{T}_{out}$ if $|\mathcal{T}_{in}| = |\mathcal{T}_{out}|$, where $|\cdot|$ denotes the dimension of a tensor; “ $\oplus$ ” means a channel-wise summarization. It means that there is no residual connection when $|\mathcal{T}_{in}| \neq |\mathcal{T}_{out}|$. The dimension difference is a consequence of the spatial change or the channel stretchiness of $\mathcal{T}_{in}$ during the learning process. Addressing this issue, we propose to make a full-residual mechanism based on the flexible property of the pointwise convolution (see Section 3.1.2 for more detail). Experimental results and evaluations in Section 4.2 have validated its efficiency in training light-weight neural networks.

3. Proposed method

In this section, an efficient FR-PDP perceptron is proposed to take advantage of the fusion of depthwise/pointwise features that are enriched by a full-residual mechanism for identity mappings. A tick-shape backbone is then introduced subject to the channel elasticity of FR-PDP blocks. Three novel concepts of constructing a light-weight network are then introduced by hooking tick-shape backbones. Hereafter, these significant proposals will be detailed.

3.1. A full-residual point-depth-point perceptron

As mentioned above, the DSC block of MobileNetV1 [18] directly exploits depthwise-based features of an input tensor, while the LiB one of MobileNetV2/V3 [19, 20] and EfficientNet [23] attempts to learn image representation by expanding and shrinking the depth channel of the tensor. In addition, just the semi-residual connection is exploited in their CNN-based architecture. To mitigate these issues, we propose an efficient FR-PDP perceptron by incorporating two components: a point-depth-point (PDP) module and a full-residual (FR) mechanism. Let us consider an input tensor $\mathcal{T} \in \mathbb{R}^{C_{in} \times H \times W}$, where C_{in} denotes a channel collection of $\mathcal{T}$; $H \times W$ means the 2D size of each channel. Hereafter, the proposed FR-PDP perceptron will be detailed as follows¹.

3.1.1. Point-depth-point features

An effective module for extracting PDP-based features is defined as a corporation of various kinds of learning operations. As intuitively shown in the light-green rectangle of Fig. 1, we firstly address $C_{in} \times 1 \times 1$ kernels ζ_j for tensor $\mathcal{T}$ in C_{in} times to obtain an output with the same #channels of $\mathcal{T}$ as

$$\text{Pw1} = \zeta_j * \mathcal{T}, \quad j = 1, \dots, C_{in} \quad (4)$$

where $\text{Pw1} \in \mathbb{R}^{C_{in} \times H \times W}$. Note that this pointwise operation will be executed with $\text{stride} = 1$ but without batch normalization and activation function. It aims to produce new point-compressed features with genuine attributes for a coming depthwise convolution, which the DSC-based features of MobileNetV1 [18] have been missing. Experimental results in Table 3 and evaluations in

¹If the convolutional attributes are not mentioned explicitly, the following properties: $\text{stride} = 1$, $\text{padding} = 0$, batch normalization [40], and ReLU activation [41] will be utilized in default for all convolutional operations. Furthermore, the biases will be ignored for simple presentation.

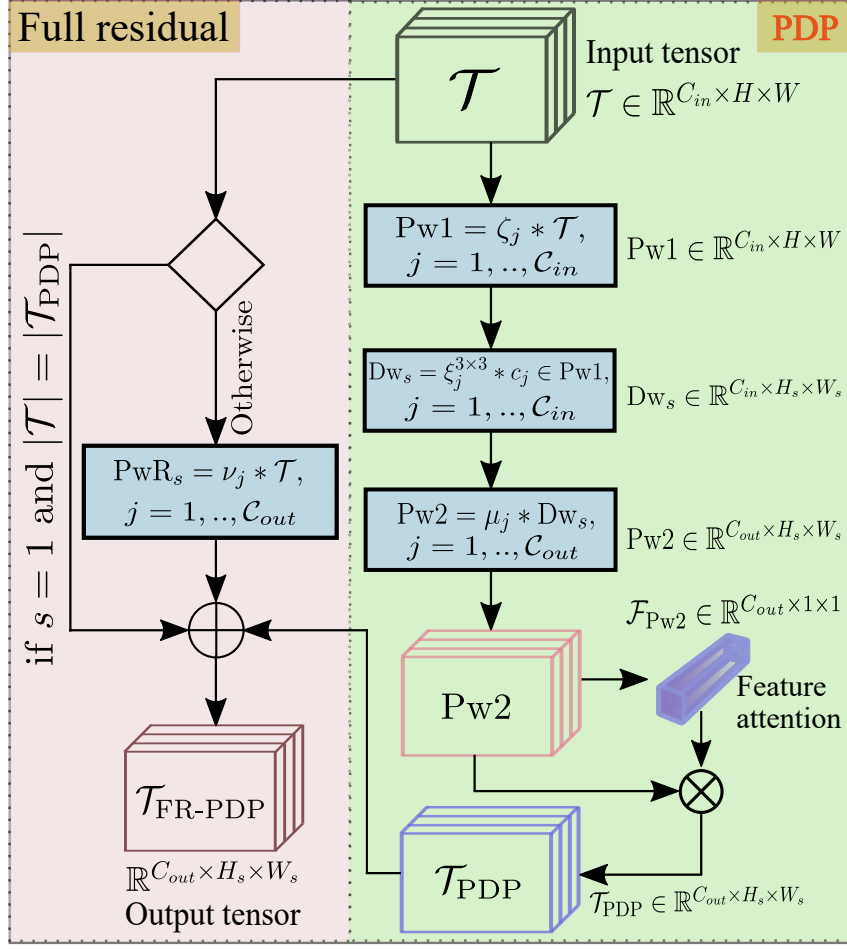


Figure 1: An intuitive illustration of the proposed FR-PDP perceptron.

Section 4.2 have proved the need for this process. As defined above, Pw1 can maintain the depth feature channels of tensor $\mathcal{T}$ due to the C_{in} -time iterations, i.e., $|\text{Pw1}| = |\mathcal{T}|$ where $|\cdot|$ returns the size of a tensor. It is different from the LiB block exploited in MobileNetV2/V3 [19, 20] and EfficientNet [23], where the #channels of an input tensor will be expanded by a pointwise convolution at the beginning of the LiB's learning stage, i.e., $|\text{Pw1}| > |\mathcal{T}|$ due to addressing C'_{in} -time iterations ($C'_{in} > C_{in}$) (see Table 1 for a brief of differences). Secondly, to learn the spatial information of the point-compressed features for image representation, we take advantage of a depthwise convolution for a light-weight number of learnable parameters as

$$\text{Dw}_s = \xi_j^{3 \times 3} * c_j \in \text{Pw1}, \quad j = 1, \dots, C_{in} \quad (5)$$

where $\text{Dw}_s \in \mathbb{R}^{C_{in} \times H_s \times W_s}$; c_j denotes a spatial channel convolved by a 3×3 kernel (i.e., $\xi_j^{3 \times 3}$) with $\text{padding} = 1$ and $\text{stride} = 1$ in normal conditions, while $\text{stride} = s$ ($s > 1$) is in some crucial circumstances for a spatial downsample to $H_s \times W_s$. It can be seen that the depthwise operation will convolve each channel c_j with a different kernel. As a result, the feature maps of Dw_s have not been incorporated to make new learning characteristics. Motivated by the DSC and LiB blocks,

Table 1: Comparing the main properties of FR-PDP with those of other tiny perceptrons.

Tiny network	Light-weight block	Residual	Other main differences
MobileNetV1 [18]	DSC [18]	No	Without point-compressed features and no feature attention
	BSCov-U [29]	No	Intra-kernel correlations for DSC blocks
MobileNetV2/V3 [19, 20]	Linear bottleneck [19]	Partly	Expanded and shrunk features
	BSCov-S [29]	Partly	Intra-kernel correlations for linear bottleneck blocks
MobileNeXt [30]	Sandglass	No	Two depthwise convolutions for more spatial information
ShuffleNetV1/V2 [21, 22]	Shuffle Unit [21]	Partly	Grouped and shuffled features
EfficientNet [23]	Linear bottleneck [19]	Partly	Expanded and shrunk features
Proposed TickNets	Proposed FR-PDP	Full	Point-compressed features and keeping depth channel cognition

we address another pointwise operation to capture the channel-wise correlation of the resultant tensor Dw_s as

$$Pw2 = \mu_j * Dw_s, \quad j = 1, \dots, C_{out} \quad (6)$$

where μ_j has the same properties as kernel ζ_j . This pointwise operation has to be iterated C_{out} times to be agreed with the #channels expected for the output tensor of the proposed PDP module. In other words, $Pw2 \in \mathbb{R}^{C_{out} \times H_s \times W_s}$. Furthermore, to enrich more robust information for image representation, we propose to take advantage of each attentive value $f_k \in \mathcal{F}_{Pw2}$ for each channel $c_k \in Pw2$ as

$$\mathcal{T}_{PDP} = c_k \otimes f_k, \quad k = 1, \dots, C_{out} \quad (7)$$

where “ $\otimes$ ” indicates channel-wise multiplication; $\mathcal{F}_{Pw2} = \{f_1, \dots, f_{C_{out}}\}$ denotes a collection of attentive light weights conducted from $Pw2$ by an attention operation. It should be noted that the attention mechanism has been exploited in several small networks, e.g., MobileNetV3 [20] and EfficientNet [23], while MobileNetV2 [19] is modified by Zhao *et al.* [36] for attentive information. However, these attentive weights were directly computed from a depthwise-based tensor whose filtered features had merely been taken into account for the training processes due to the inherent characteristic of the depthwise convolution. Different from them, we propose to take the pointwise-based features for the attention computation, i.e., $\mathcal{F}_{Pw2}$, where the new informative features have been made from the depthwise-based tensor (i.e., Dw_s). For objective assessments, we simply address the common attention module, i.e., squeeze and excitation (SE) [42] with the default value of reduction ratio (i.e., $r = 16$).

3.1.2. A full-residual mechanism for light-weight perceptron

As well known, the residual mechanism [25] for a CNN-based block is generally an element-wise summarization of an input tensor and its output features that have been learned through several neural layers. In several existing small networks, the residual mechanism has not been exploited, i.e., MobileNetV1 [18]. If have, it is just utilized partly (semi-residual) in their perceptron modules [19, 20, 23, 21]. This is due to *i*) the change of spatial information of an input tensor when applying *stride* > 1 for the depthwise convolution (i.e., *stride* = 2 in those modules), and *ii*) the stretchiness of the channels during the inner learning process. To deal with these shortcomings, we propose a full-residual mechanism thanks to the flexibility of a pointwise operation. It aims to transform an input tensor $\mathcal{T} \in \mathbb{R}^{C_{in} \times H \times W}$ so that the dimension of its resultant volume is identical to that of the output of a light-weight perceptron for identity mappings. Indeed, the pointwise

operation with $stride = s$, ($s > 1$) for this determination is defined as

$$\text{PwR}_s = \nu_j * \mathcal{T}, \quad j = 1, \dots, C_{out} \quad (8)$$

where ν_j denotes a $C_{in} \times 1 \times 1$ kernel, $\text{PwR}_s \in \mathbb{R}^{C_{out} \times H_s \times W_s}$. It should be noted that the practice stride for PwR_s will be addressed subject to the backbone of a specific network.

Taking into account the proposed full-residual mechanism for the above PDP block, we can make a full-skip connection for tensor $\mathcal{T}_{\text{PDP}}$ as.

$$\mathcal{T}_{\text{FR-PDP}} = \begin{cases} \mathcal{T} \oplus \mathcal{T}_{\text{PDP}}, & \text{if } s = 1 \text{ and } |\mathcal{T}| = |\mathcal{T}_{\text{PDP}}|, \\ \text{PwR}_s \oplus \mathcal{T}_{\text{PDP}}, & \text{otherwise.} \end{cases} \quad (9)$$

where $\mathcal{T}_{\text{FR-PDP}} \in \mathbb{R}^{C_{out} \times H_s \times W_s}$; $|\cdot|$ returns the dimension of a tensor; “ $\oplus$ ” denotes an operator of channel-wise summarization. The stride value for computing PwR_s in this case is in accordance with that for the computation of Dw_s . An intuitive view of the full-residual operation is illustrated in the light-pink rectangle of Fig. 1. As constructed above, the proposed FR-PDP block would take two kinds of residual features for boosting performance: *i*) the conventional residual features in the case of $stride = 1$ and the dimension of input and output PDP tensors are identical, i.e., $|\mathcal{T}| = |\mathcal{T}_{\text{PDP}}|$; *ii*) the PwR_s -based ones in other circumstances. Furthermore, we also deploy this proposed full-residual mechanism to light-weight perceptrons of several small CNN-based networks [18, 19, 20, 30] to investigate further effectiveness. Experimental results have indicated the efficacy of the proposed full-residual mechanism. Please refer to Section 4.2 for thorough evaluations.

Based on the above constructions, it can shortly conduct the following distinctions of FR-PDP to enhance the learning process for image representation in comparison with other light-weight blocks (see Table 1 for a brief of the major differences).

- FR-PDP produces point-compressed features and maintains the depth channel cognition of an input tensor within the representative perceptron, while the channel deformation is addressed in the LiBs of MobileNetV2/V3 [19, 20] and EfficientNet [23].
- Addressing the pointwise-based attentive features allows FR-PDP to enrich more robust attentive information since the pointwise convolution can point out new features for the learning process. Otherwise, the depthwise-based features in several light-weight networks [20, 23, 36] would be not.
- A full-residual mechanism for a light-weight perceptron module is proposed to deal with the limitation of skip connections due to the spatial downsampling operations.
- The proposed full-residual mechanism allows FR-PDP to exploit diverse kinds of skip-connection features for further improvement: those from the conventional residual computation and those from PwR_s .

3.2. Tick-shape networks of FR-PDP blocks

As well known, small CNN-based networks [18, 19, 23, 20, 21] often have a “plumb” backbone, where the number of output channels would be steadily increased after each light-weight block is executed. Different from them, we propose to design disparate networks where a period of channel elasticity is in accordance with a tick-shape model as follows.

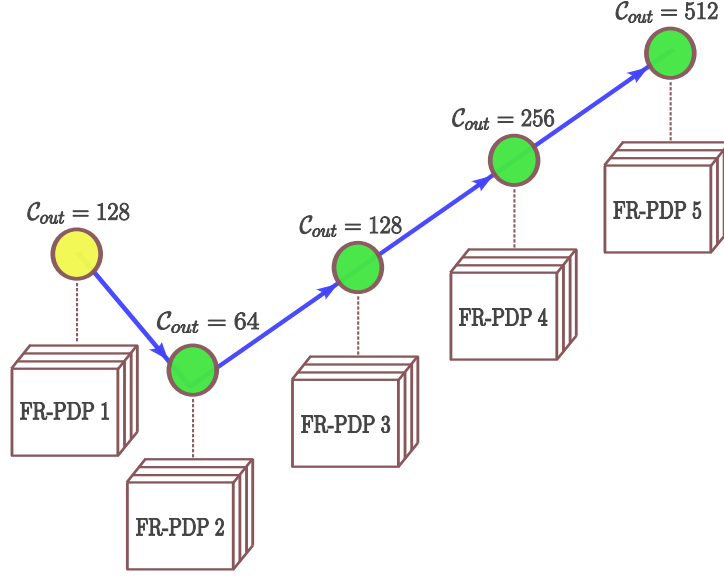


Figure 2: TickNet-basic: A basic tick-shape network of FR-PDP blocks, where circular nodes mean that FR-PDP is handled for an output of C_{out} channels; Green nodes are for executing Dw_s of FR-PDP with stride $s = 2$, while the yellow one is with $s = 1$; Arrows indicate the direction of learning processes.

3.2.1. A basic tick-shape backbone

We suppose that a tensor $\mathcal{T} \in \mathbb{R}^{C_{in} \times H \times W}$ represents an input image. A basic tick-shape backbone of FR-PDP blocks is designed as

$$\text{Tick_B}(\mathcal{T}) = \Psi_s^{512}(\Psi_s^{256}(\Psi_s^{128}(\Psi_s^{64}(\Psi_s^{128}(\mathcal{T})))))) \quad (10)$$

where $\Psi_s^{C_{out}}(.)$ is an execution of a FR-PDP block for an output with C_{out} channels using a spatial downsampling operation with $stride = s$ if $s > 1$. For instance, $\Psi_2^{128}(\mathcal{T})$ means that FR-PDP will handle a given tensor $\mathcal{T}$ with $stride = 2$ for an output of $C_{out} = 128$ channels. Fig. 2 illustrates an intuitive view of the basic tick-shape backbone, where the circular nodes mean that FR-PDP will be carried out for a resultant tensor with C_{out} channels subject to stride $s = 2$ at the circular green nodes or $s = 1$ at the yellow one.

3.2.2. Efficient TickNets based on tick-shape backbones

The smallest network of FR-PDP blocks, named TickNet-basic, is proposed to be identical to one basic tick-shape backbone for learning 224×224 images with spatial downsampling operations, as shown at the green nodes in Fig. 2. It should be noted that for learning more spatial information in smaller images (i.e., those with the size of 32×32 in dataset CIFAR10/100 [43]), the second FR-PDP node will be handled with stride $s = 1$, i.e., $\Psi_{s=1}^{64}(\Psi_{s=1}^{128}(\mathcal{T}))$ (refer to Eq. (10)). In other words, the node with $C_{out} = 64$ in Fig. 2 will be decorated in yellow color in this circumstance, instead of the green one (see Table 2 for the architecture of TickNet-basic in detail).

Thanks to the proposed concept of channel elasticity for a backbone, we introduce two more light-weight networks for the learning representation of 224×224 images by hooking together

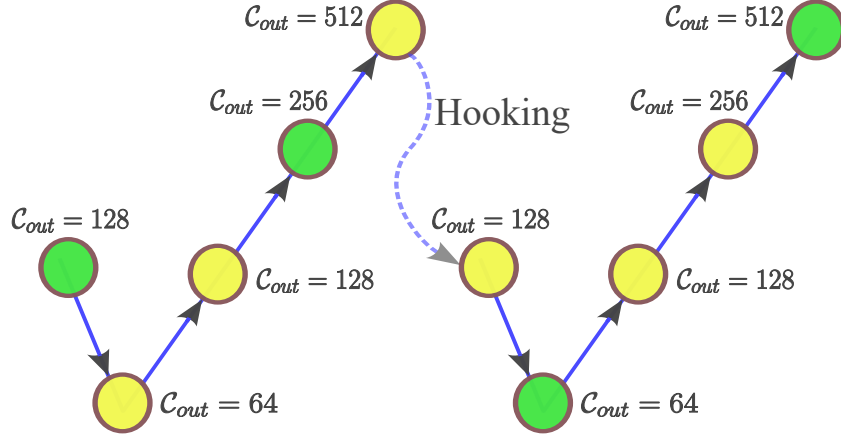


Figure 3: Illustration of TickNet-small with two basic tick-shape backbones.

several basic tick-shape backbones. TickNet-small is formed by hooking two basic backbones as

$$\text{TickNet-small}(\mathcal{T}) = \text{Tick_B}(\text{Tick_B}(\mathcal{T})) \quad (11)$$

So TickNet-small would include 10 FR-PDP blocks (see Table 2 for its architecture in more detail).

Fig. 3 shows an intuitive view of hooking together two tick-shape models with a distribution of green nodes where a FR-PDP block will go with $\text{stride} = 2$. Similarly, a deeper tick-shape network (named TickNet-large) consists of 15 FR-PDPs structured by hooking three basic backbones as

$$\text{TickNet-large}(\mathcal{T}) = \text{Tick_B}(\text{Tick_B}(\text{Tick_B}(\mathcal{T}))) \quad (12)$$

Table 2 shows the full structure of TickNet-large, while Fig. 4 indicates which nodes are mentioned for its spatial downsampling operations (i.e., $s = 2$). Note that the first node of TickNet-small/large (i.e., FR-PDP-1 in Table 2) is decorated for a stride $s = 1$ to capture more spatial features in learning small images (e.g., 32×32 images in CIFAR10/100). The related code and the proposed tick-shape models can be downloaded at <https://github.com/nttbdrk25/TickNets>.

3.2.3. Computational complexity of TickNets

For a simple estimation of memory footprint, we would like to disregard the biases and the change of spatial dimension $H \times W$ in the learning processes. As defined in Section 3.1, the proposed light-weight FR-PDP block consists of four main CNN-based computations because the complexity of a SE attention block [42] is negligible. The number of learnable parameters will be approximately estimated subject to the input and output channels as follows: $Q_{Pw1} = C_{in} \times C_{in}$; $Q_{Dw_s} = 3 \times 3 \times C_{in}$; $Q_{Pw2} = Q_{PwR_s} = C_{in} \times C_{out}$. We assume $C_{in} \approx C_{out}$ for a rough approximation. So #parameters of a FR-PDP block are computed as

$$\begin{aligned} Q_{\text{FR-PDP}} &\approx Q_{Pw1} + Q_{Dw_s} + Q_{Pw2} + Q_{PwR_s} + Q_{\mathcal{F}} \\ &\approx 3C_{out}^2 + 9C_{out} \end{aligned} \quad (13)$$

Similarly, it can be deduced that the complexity of a depthwise separable convolution block in MobileNetV1 is estimated as $Q_{\text{DSC}} \approx C_{out}^2 + 9C_{out}$, while that of a linear bottleneck block in MobileNetV2/V3 and EfficientNet is $Q_{\text{LiB}} \approx 2C_{out}^2 + 9C_{out}$.

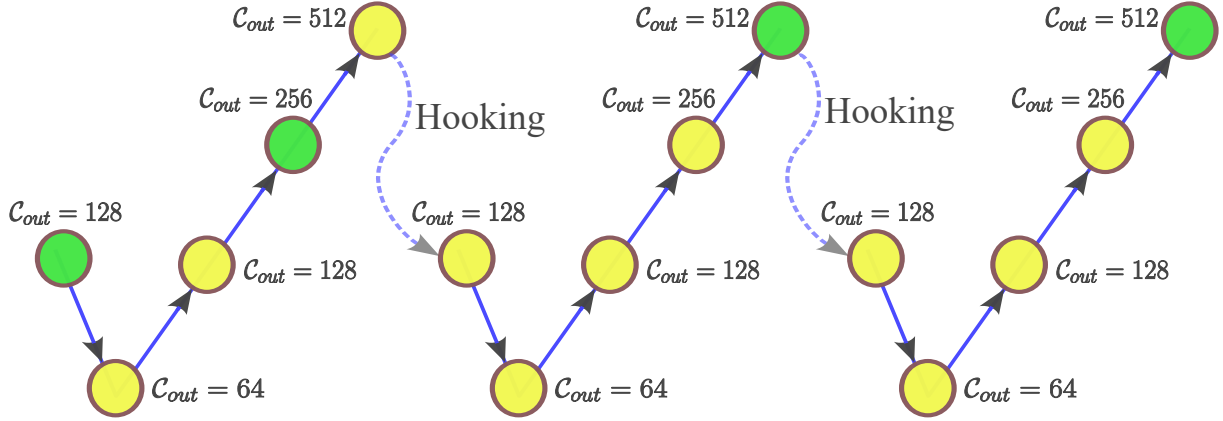


Figure 4: Illustration of TickNet-large with three basic tick-shape backbones.

As designed in Section 3.2.2, the complexity of TickNets would be multiple of $Q_{\text{FR-PDP}}$ subject to how many FR-PDP blocks have been addressed in the whole architecture. For instance, $Q_{\text{TickNet-basic}} \approx 5Q_{\text{FR-PDP}}$ since TickNet-basic has 5 FR-PDP blocks (see Table 2). Although $Q_{\text{FR-PDP}}$ raised about 50% compared to Q_{LiB} and two times versus Q_{DSC} , the obtained TickNets of FR-PDPs have less learnable parameters in general. This is thanks to the novel concept of a tick-shape backbone that allows us to design shallow FR-PDP-based networks with good performance. For instance, TickNet-small with 10 FR-PDP blocks has 3M learnable parameters on ImageNet-1k versus 3.51M of MobileNetV2. It is because MobileNetV2 goes with a deeper architecture exploiting 17 LiB perceptrons. Despite of $Q_{\text{FR-PDP}} = 2Q_{\text{DSC}}$, TickNet-large with 15 FR-PDPs just has 3.92M compared to 4.23M of MobileNetV1 with 13 DSC blocks. The sharp increase of #parameters is caused by the MobileNetV1’s model addressing depth channels in its last layers. In the meantime, the proposed TickNets are at a smaller cost and obtain better results (Refer to Tables 3, 6, and 7 for thorough comparison).

4. Experiments and evaluations

4.1. Datasets and learning settings

In order to comprehensively evaluate the efficiency of TickNets with FR-PDP blocks, the proposed networks (i.e., TickNet-basic, TickNet-small, and TickNet-large) will be trained on benchmark datasets for classifying object and scene images. For objective evaluations, the learning process utilizes the popular learning settings as a SGD optimizer with momentum = 0.9; initial learning-rate $lr = 0.1$; weight-decay = 10^{-4} . Other learning parameters will be specified subject to the properties of each dataset for objective evaluation. It should be noted that if the expansion coefficient is not mentioned explicitly, the typical one (i.e., *width-multiplier* = x1.0) will be utilized for MobileNet-based networks [18, 19, 20, 30, 29, 44].

CIFAR [43] is compiled into two subsets: CIFAR-10 includes 60k 32×32 color images which are split into 10 categories with 50k images for the training set and 10k images for the testing one; CIFAR-100 is like CIFAR-10, except 600 images are schemed for each class (i.e., 100 in total). Per class, 500 images are used for training and 100 for testing. As the learning settings addressed

Table 2: Overall architectures of TickNets structured by FR-PDP blocks for learning 224×224 images.

Layer	TickNet-basic		TickNet-small		TickNet-large		C_{out}
	Stride	Output $H \times W$	Stride	Output $H \times W$	Stride	Output $H \times W$	
Input image	-	224×224	-	224×224	-	224×224	3
Initial_conv3x3	2	112×112	2	112×112	2	112×112	32
FR-PDP-1	1	112×112	2	56×56	2	56×56	128
FR-PDP-2	2	56×56	1	56×56	1	56×56	64
FR-PDP-3	2	28×28	1	56×56	1	56×56	128
FR-PDP-4	2	14×14	2	28×28	2	28×28	256
FR-PDP-5	2	7×7	1	28×28	1	28×28	512
FR-PDP-6	-	-	1	28×28	1	28×28	128
FR-PDP-7	-	-	2	14×14	1	28×28	64
FR-PDP-8	-	-	1	14×14	1	28×28	128
FR-PDP-9	-	-	1	14×14	1	28×28	256
FR-PDP-10	-	-	2	7×7	2	14×14	512
FR-PDP-11	-	-	-	-	1	14×14	128
FR-PDP-12	-	-	-	-	1	14×14	64
FR-PDP-13	-	-	-	-	1	14×14	128
FR-PDP-14	-	-	-	-	1	14×14	256
FR-PDP-15	-	-	-	-	2	7×7	512
Final_conv1x1	1	7×7	1	7×7	1	7×7	1024
AvgPooling							1024
Fully connected							1000

Note: Initial_conv3x3 is a standard 2D convolution with a 3×3 kernel and stride $s = 2$. Final_conv1x1 is a pointwise convolution with $s = 1$. “-” means “not available”. Note that $s = 1$ will be applied to the first two blocks FR-PDP-1 and FR-PDP-2 of TickNets for representing 32×32 images in CIFAR10/100 [43].

in [25, 45, 29] for these subsets, TickNets will be trained in 200 epochs with a batch size of 128 images. The decay of learning rate $lr = 0.1$ is a factor of 0.1 at epochs: 100, 150, and 180. The augmenting techniques in [25, 45], i.e., horizontal flips and shifts by up to 4px, are taken into account for mitigating the overfitting problem. Furthermore, for objective evaluations, the first sandglass block of MobileNeXt [30] will be modified so that it is executed with stride $s = 1$ to capture more spatial information in the case of learning the 32×32 images of CIFAR-10/100.

ImageNet-1k [46] consists of 1000 categories of over 1.3M annotated images. Therein, over 1.28M are utilized for the training stage and 50k for the validating one. Subject to the learning settings in the previous works [47, 48, 29], TickNets will be trained in 100 epochs with a batch size of 256 samples. The decay of learning rate $lr = 0.1$ is a factor of 0.1 at epochs: 30, 60, and 90. As usual, the augmenting transforms will be used for the learning process: scaling the original images into 256×256 ; randomly cropping the output for 224×224 images; and horizontal flips. Furthermore, due to the largeness of ImageNet-1k, several ImageNet’s subsets of miscellaneous

Table 3: Top-1 performance (%) of semi/full-residual PDP blocks.

	Semi-residual (SR-PDP)					Full-residual (FR-PDP)				
	Stanford Dogs			ImageNet-100		Stanford Dogs			ImageNet-100	
Backbone	Top-1	#Paras	FLOPs	Top-1	#Paras	Top-1	#Paras	FLOPs	Top-1	#Paras
MobileNetV1 [18]	60.07	6.61M	0.61G	82.98	6.59M	60.71	9.76M	0.68G	83.00	9.74M
MobileNetV2 [19]	48.46	0.90M	0.05G	74.60	0.87M	50.90	1.06M	0.06G	76.26	1.04M
MobileNetV3 [20]	46.89	1.80M	0.04G	73.72	1.77M	51.01	1.92M	0.04G	75.38	1.89M
MobileNeXt [30]	56.01	11.92M	0.56G	83.48	11.90M	56.81	17.67M	0.67G	84.32	17.65M
ShuffleNetV2 [22]	55.27	3.15M	0.18G	78.30	3.13M	55.95	4.36M	0.19G	79.86	4.34M
TickNet-basic	54.48	0.99M	0.21G	78.26	0.97M	54.63	1.18M	0.25G	78.82	1.16M
TickNet-small	56.79	1.66M	0.37G	80.26	1.64M	63.50	2.10M	0.50G	83.28	2.08M
TickNet-large	62.91	2.34M	0.47G	81.88	2.32M	64.14	3.02M	0.64G	84.36	3.00M

Note: #Paras denotes a number of learnable parameters computed by using tool *ptflops* [53].

object images were retrieved as follows. Tian *et al.* [49] formed ImageNet-100 to quickly validate the performance of contrastive learning networks in image classification. It has 100 classes which were randomly selected from those of ImageNet. Meantime, Nguyen *et al.* [50] presented more challenging subsets (ReIN^Ω) with small numbers of classes extracted by addressing statistical computation subject to the validation outcomes of a pre-trained network. We would like to take into account ImageNet-100 and the three most challenging sets of ReIN^Ω (i.e., ReIN³⁰, ReIN⁵⁰, and ReIN¹⁰⁰) to validate the ability of TickNets in comparison with other light-weight models. The same learning settings on ImageNet-1k will be used for objective comparisons.

Stanford Dogs [51] is an object-species dataset constructed by only allocating ImageNet’s images of species dogs. It consists of over 20k instances split into 120 species of dogs: 12000 images for training and 8580 for validating. Following the learning settings in [29], TickNets will be trained in 200 epochs with a batch size of 64 samples. The decay of learning rate $lr = 0.1$ is a factor of 0.1 at epochs: 100, 150, and 180. Augmenting transforms will be located the same as done on ImageNet-1k along with color jitter [8].

Places365-Standard [52] consists of 365 categories. Each class has from 3068 to 5000 scene images for the training stage (i.e., over 1.8M images in total), while 36500 images are addressed for the validating stage with 100 scene images per class. Nguyen *et al.* [50] rescaled this dataset for two more challenging subsets: RePL³⁰ and RePL⁵⁰. The same learning settings on ImageNet-1k will be used to train TickNets on Places365-Standard and its subsets for objective comparisons.

4.2. Analyzing efficacy of FR-PDP-based features

As defined in Section 3.1, the proposed FR-PDP block is structured by two main ingredients: full-residual (FR) and point-depth-point (PDP). In this section, we will investigate the efficacy of FR-PDP and its components on several popular light-weight networks. To this end, we deploy both PDP and FR-PDP components for TickNets and other MobileNet-based networks. This means that the DSC and LiB blocks in MobileNets [18, 19, 20], and the sandglass blocks in MobileNeXt [30] will be replaced by either semi-residual PDP blocks (SR-PDPs) or FR-PDPs, correspondingly.

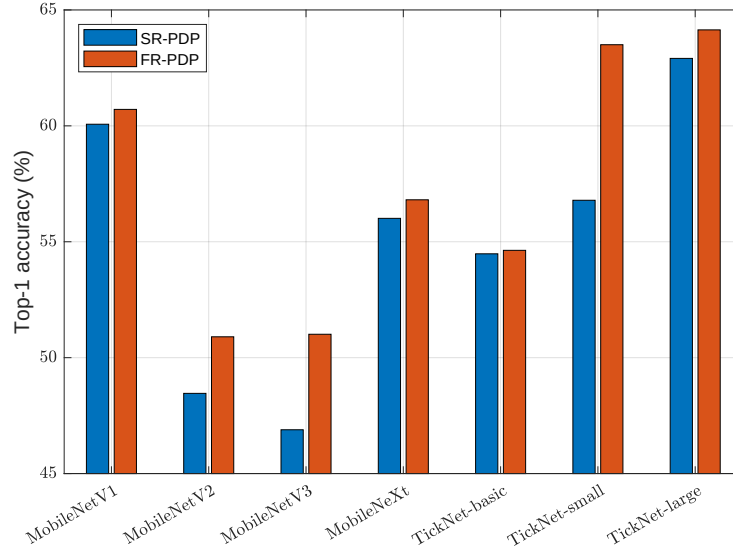


Figure 5: The efficacy of FR-PDPs compared to SR-PDPs on Stanford Dogs.

It can be observed in Table 3 that the FR-PDP-based networks obtained better performance in image representation, particularly the noteworthy contribution of FR-PDPs in boosting the performance of TickNet-small and MobileNetV2/V3 (see Fig. 5). Concretely, MobileNetV1 [18] with the SR-PDP blocks achieved 60.07% and 82.98% on Stanford Dogs and ImageNet-100, respectively about 8.5% and 6% higher than those of MobileNetV1 using DSC blocks (51.6% and 77.01%, see Tables 6 and 5). However, the learnable parameters have been raised doubly (e.g., 6.59M versus 3.33M on Stanford Dogs) due to the depth channels at the last layers of MobileNetV1. MobileNeXt [30] with FR-PDPs obtained better results than with SR-PDPs but the sharp increase of learnable parameters (e.g., 17.65M versus 11.9M on ImageNet-100). In terms of MobileNetV2/V3² with SR-PDPs and FR-PDPs, their performance has decreased in comparison with that of the LiB-based models (see Tables 3, 6, and 5). This would be because of the shallow layers of their backbone. It means that the channel expansion of the LiB block plays an important role in the entire learning process of MobileNetV2/V3 since it makes MobileNetV2/V3 more complex with much more learnable parameters (from 2.5 M to 3.9 M) justifying those differences in performance. In another aspect, the results of MobileNetV2/V3 using SR-PDPs and FR-PDPs have also verified the prominence of our pointwise-based full-residual mechanism for light-weight CNN-based modules (see Fig. 5). For instance, the FR-PDPs boosted the performance of MobileNetV3 up to 4%, i.e., from 46.89% to 51.01% with the slight increase of #parameters, i.e., from 1.8M to 1.92M.

In respect of the influence of the proposed full-residual mechanism on TickNets, Table 3 also shows that the performance of TickNets has been boosted as expected. For instance, TickNet-small with FR-PDP obtained 63.5% on Stanford Dogs, about 6.5% higher than TickNet-small (56.79%) with the semi-residual features, while the growth of #parameters is just from 1.66M

²It should be noted that the performance of MobileNetV3 [20] addressed in this work for comparison is resulted by its largest version, i.e., “version=large” and “use_lightweight_head=False”.

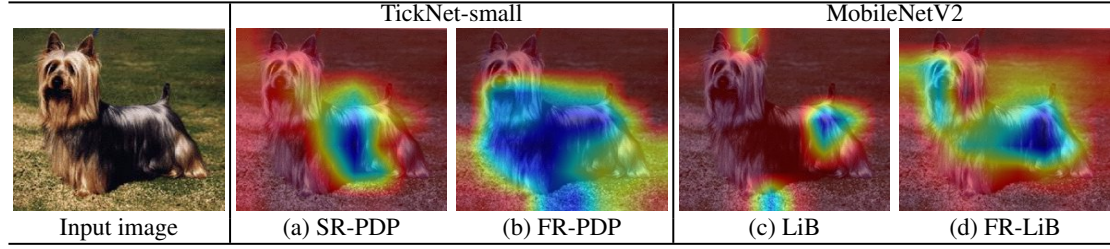


Figure 6: Efficacy of full-residual (FR) compared to that of semi-residual (SR) on the LiB and PDP blocks for learning an Australian_terrier dog of Stanford Dogs. Visualizations are drawn out by using Grad-CAM [54].

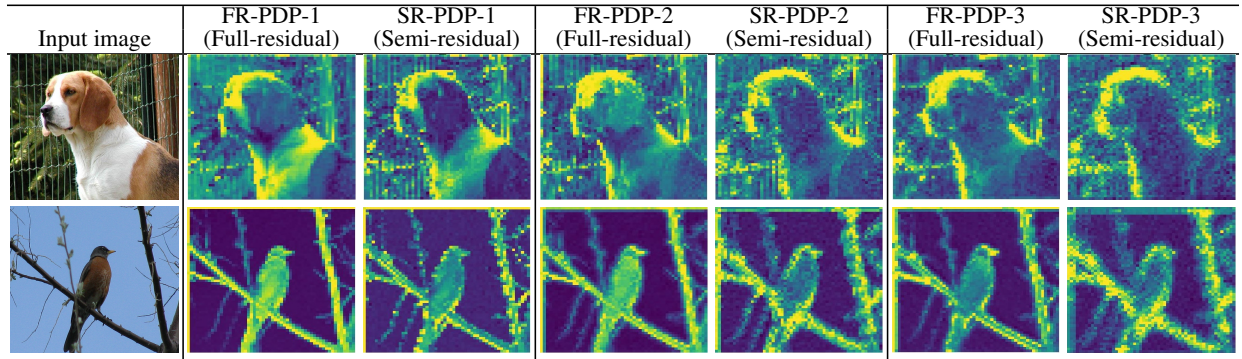


Figure 7: Efficacy of full-residual (FR) compared to semi-residual (SR) in the first three PDP-based blocks of TickNet-small. Visualizations are for the feature maps of a beagle-dog image and an adult-robin image.

to 2.1M. This advance would be thanks to more spatial information of the PDP-based features enhanced by the full-residual mechanism. Indeed, Fig. 6(a, b) indicates that most body parts of an Australian_terrier-dog have been addressed in the learning process of the full-residual mechanism, while Fig. 7 visualizes the solid information of spatial patterns of a beagle dog and an adult robin that are represented through the first three PDP-based blocks of TickNet-small: the full-residual blocks (i.e., FR-PDP-1, FR-PDP-2, and FR-PDP-3) in pairwise comparison with the semi-residual ones (i.e., SR-PDP-1, SR-PDP-2, and SR-PDP-3).

Furthermore, to verify the competence of the full-residual mechanism, we would like to deploy it for the DSC and LiB blocks of MobileNets [18, 19, 20] as well as for the perceptron of other light-weight models [22, 30, 23]. It can be observed in Table 4 that these full-residual MobileNets obtained better results than the original ones, i.e., MobileNetV1 without residual features or MobileNetV2/V3 with an incomplete residual mechanism. Particularly, the performance of EfficientNet-B0 [23] was boosted up to about 8% (from 54.7% to 62.64%) on Stanford Dogs, while the performance of MobileNetV2 was enhanced up to 5%, from 54.1% to 59.09%, also see Fig. 8 for a visual view of this advance. It is thanks to more spatial information enriched by the pointwise-based full-residual features (see Fig. 6(c) versus Fig. 6(d)). In short, the aforementioned assessments have substantiated the efficacy of the proposed full-residual mechanism along with PDP-based features for the light-weight CNN-based networks.

Table 4: Top-1 rates (%) of light-weight networks taking semi/full residual mechanism.

Architecture	Without residual or semi-residual (SR)			Proposed full-residual (FR)		
	CIFAR-10	Stanford Dogs	ImageNet-100	CIFAR-10	Stanford Dogs	ImageNet-100
MobileNetV1 [18]	93.76	53.58	77.01	93.80	53.60	78.82
MobileNetV2 [19]	94.06	54.10	77.87	94.27	59.09	80.56
MobileNetV3 [20]	94.08	50.76	77.09	94.14	53.63	79.22
MobileNeXt [30]	93.80	53.95	81.44	94.41	54.95	83.48
ShuffleNetV1 [21]	92.29	51.02	75.42	92.70	54.87	77.76
ShuffleNetV2 [22]	93.93	54.29	79.22	94.07	56.04	79.50
EfficientNet-B0 [23]	-	54.70	80.74	-	62.64	81.00

Note: “-” means “not available” since EfficientNet [23] has not been introduced for learning 32×32 images.

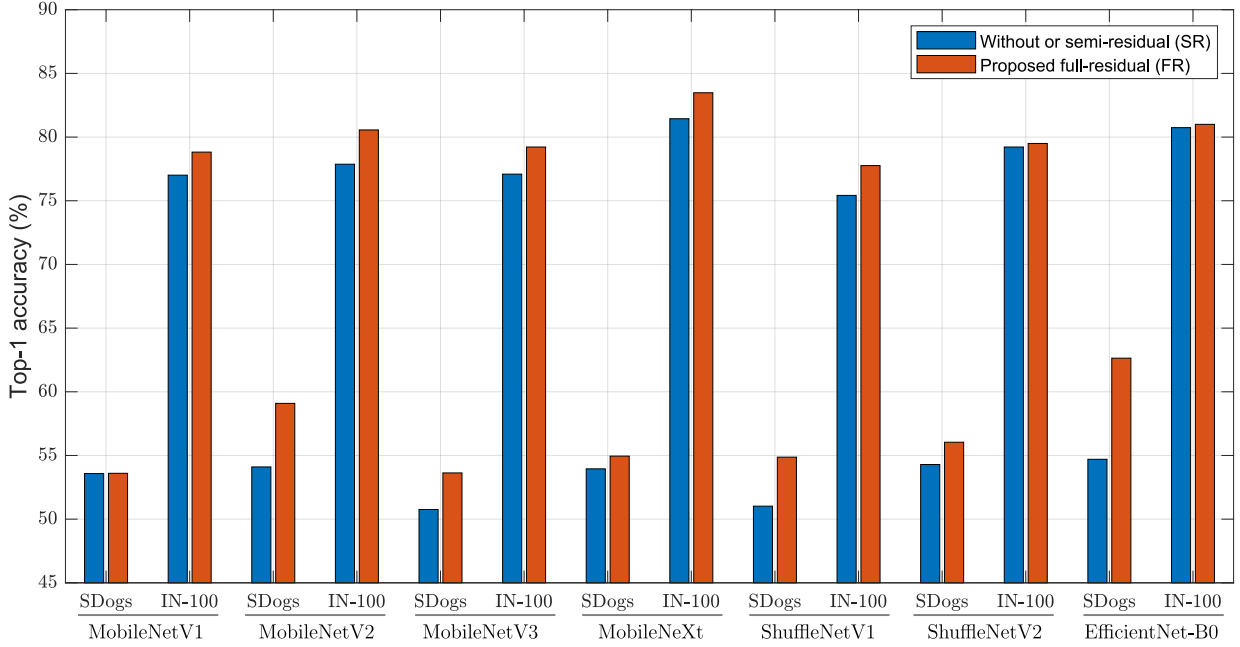


Figure 8: The effectiveness of the proposed full-residual mechanism for MobileNets and other light-weight networks on Stanford Dogs (SDogs) and ImageNet-100 (IN-100).

4.3. Performance assessment of tick-shape networks

It can be observed in Table 3 that the networks with tick-shape backbones achieve better results than the MobileNets’ backbones with the steady-increased #channels. Indeed, TickNet-small obtained 63.5% on Stanford Dogs with only 2.1M parameters, about 2.8% higher than MobileNetV1 [18] (60.71% with up to 4.6 times of #parameters, i.e., 9.76M). In the meanwhile, MobileNetV2 [19] using FR-PDP blocks has the same #parameters as TickNet-basic but about 3.7% lower performance, particularly, with a double complexity in the case of MobileNetV3 [20]. On the other hand, the results of TickNets are also much better than MobileNets’ with the original DSC and LiB blocks (see Tables 6 and 5). Those factors have proved the efficacy of the channel elasticity against the steady growth of channels that were conventionally addressed in the backbone of tiny

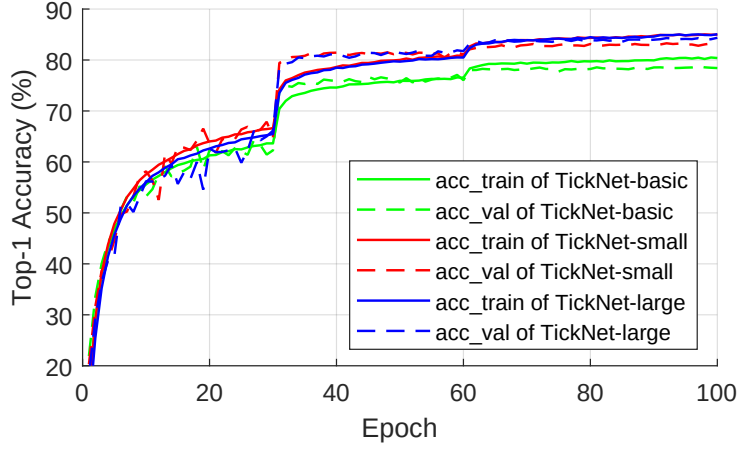


Figure 9: Training and validating plots of TickNets on ImageNet-100.

networks. Also, those have consolidated the prominence of the FR-PDP-based features.

Integrating together tick-shape backbones has made the resultant network more robust for image representation. The experimental results of TickNets in Tables 6, 5, and 7 indicate that the performance would be improved subject to the quantity of tick-shape backbones involved. For instance, TickNet-basic with one tick-shape backbone (see Fig. 2) just obtained 54.63% and 78.82% on Stanford Dogs and ImageNet-100, respectively. These rates are $\sim 9\%$ and $\sim 4.5\%$ lower than TickNet-small (63.5% and 83.28%) with two integrated backbones (see Fig. 3). For an intuitive view, it can be observed in Fig. 9 that TickNet-small has better learning ability than TickNet-basic in the image description. Meantime, the performance of TickNet-large with three hooked backbones (see Fig. 4) is just a little higher than TickNet-small's (see Fig. 9 for training and validating plots). Indeed, TickNet-large obtained 64.14% and 84.36% on Stanford Dogs and ImageNet-100 but with a sharp increase ($\times 1.5$) of #parameters (i.e., 3M, see Table 3). As a visual concentrating instance in Fig. 10, it can be seen that TickNet-large has a better concentration on the solid spatial information of geese, one of the crucial attributes to recognize which kinds of the concerning objects. In addition, the above assessments are in accordance with the learning results of TickNets on other datasets, i.e., CIFAR10/100 in Table 6; ImageNet-1k, Places365, and their challenging subsets in Tables 5 and 7.

Consequently, based on the aforementioned assessments, TickNet-small should be recommended for real applications in mobile devices as well as embedded systems with the limitation of computing resources. This assertion is thanks to the good performance of TickNet-small with a low quantity of light-weight parameters for image representation, while it is little lower performance in comparison with that of TickNet-large (see Fig. 11). In the case of higher-accuracy requirements, TickNet-large would be a potential solution in practice. Indeed, it can be considered as an alternative model for real applications such as coal-gangue recognition in the coal mining industry [55], fruit classification [56], bamboo stick counting [57], unmanned aerial vehicles (UAVs) [12, 58, 59].

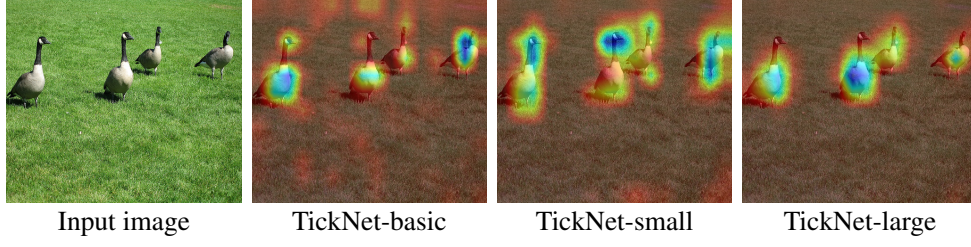


Figure 10: Efficient concentration of TickNets on representing the spatial features of geese. These visualizations are formed by using Grad-CAM [54].

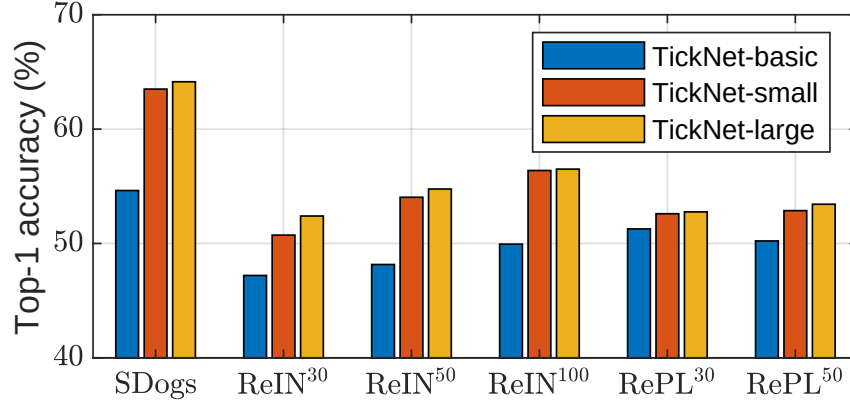


Figure 11: Performances of TickNets on Stanford Dogs (SDogs) and challenging subsets (ReIN) and (RePL) of ImageNet-1k and Places365.

Table 5: Top-1 performance (%) on subsets of ImageNet-1k and Places365.

Architecture	ReIN ³⁰	ReIN ⁵⁰	ReIN ¹⁰⁰	ImageNet-100	RePL ³⁰	RePL ⁵⁰
MobileNetV1 [18]	46.53	50.08	52.48	77.01	51.53	50.28
MobileNetV2 [19]	47.47	49.08	53.74	77.87	52.67	52.02
MobileNetV3 [20]	49.87	51.68	54.06	77.60	52.33	51.68
MobileNeXt [30]	48.80	52.76	52.96	81.44	51.24	52.17
ShuffleNetV2 [22]	48.87	49.52	51.38	79.22	50.97	51.12
ResNet18 [25]	47.67	50.08	53.38	81.40	51.30	52.28
TickNet-basic	47.20	48.16	49.94	78.82	51.27	50.22
TickNet-small	50.73	54.04	56.38	83.28	52.60	52.87
TickNet-large	52.40	54.76	56.50	84.36	52.76	53.43

Note: Results of other networks are reported in [50].

4.4. Comprehensive comparison with state of the art

4.4.1. Classification on CIFAR-10/100 and Stanford Dogs

It can be observed from Table 6 that TickNets have good performance with the lowest number of learnable parameters in comparison with other light-weight CNN-based networks. Concretely,

Table 6: Top1 performance (%) on CIFAR-10/100 and Stanford Dogs.

	CIFAR-10		CIFAR-100		Stanford Dogs		
Architecture	Top-1	#Paras	Top-1	#Paras	Top-1	#Paras	FLOPs
MobileNetV1 [18]	93.76	3.22M	74.27	3.31M	51.60	3.33M	0.29G
MobileNetV2 [19]	93.60	2.24M	74.90	2.35M	56.01	2.37M	0.17G
MobileNetV3 [20]	93.70	2.98M	75.01	3.07M	54.90	3.09M	0.12G
MobileNeXt [30]	93.80	2.27M	74.30	2.38M	53.95	2.41M	0.17G
V1-BSEConv-U [29]	94.30	3.22M	75.70	3.31M	59.10	3.33M	0.44G
V2-BSEConv-S [29]	94.20	2.24M	75.80	2.35M	60.10	2.38M	0.17G
V3-BSEConv-S [29]	94.60	2.98M	77.70	3.07M	60.00	3.09M	0.12G
V1-GDF [44]	93.87	3.28M	74.48	3.37M	54.90	3.39M	0.31G
V2-GDF [44]	93.66	2.32M	75.11	2.43M	57.92	2.46M	0.18G
ShuffleNetV1 [21]	92.29	0.91M	70.06	1.00M	51.02	1.02M	0.07G
ShuffleNetV2 [22]	93.93	2.49M	74.53	2.58M	54.29	1.38M	0.08G
ResNet18 [25]	93.02	11.17M	75.10	11.20M	52.57	11.24M	0.91G
ResNet18 [25] + SE [42]	-	-	76.44	11.40M	53.48	11.33M	0.91G
AugShuffleNet [60]	93.63	1.21M	74.07	1.30M	-	-	-
SqueezeNet [61]	-	-	69.41	0.78M	-	-	-
DenseNet-121 [62]	94.23	6.96M	76.21	7.06M	56.90	7.08M	1.44G
MnasNet [24]	-	-	-	-	54.80	3.26M	0.17G
EfficientNet-B0 [23]	-	-	-	-	54.70	4.16M	0.20G
TickNet-basic	92.72	1.07M	71.91	1.16M	54.63	1.18M	0.25G
TickNet-small	94.57	1.99M	76.83	2.08M	63.50	2.10M	0.50G
TickNet-large	95.01	2.91M	77.23	3.00M	64.14	3.02M	0.64G

Note: “-” means “not available” or “not to be reported”. #Paras is a number of learnable parameters computed by using tool *ptflops* [53].

TickNet-large obtained 95.01% on CIFAR-10 and 77.23% on CIFAR-100 with about 3M parameters, while TickNet-small achieved 94.57% and 76.83% with just 2M, respectively. It should be noted that V3-BSEConv-S (i.e., LiBs of MobileNetV3 [20] replaced by BSEConv-S blocks [29]) obtained the same performance on CIFAR-100 (77.7%) but just 94.6% on CIFAR-10. In the meantime, the rate of ResNet18 [25] + SE [42] is 76.44% on CIFAR-100, the same as TickNet-small’s but it needs up to 5 times of #parameters (11.4M). DenseNet-121 [62] obtained 76.21% with 7.06M, up to 3.5x versus TickNet-small’s (2.08M). In terms of image classification on Stanford Dogs, the performance of TickNet-large (64.14%, 3.02M, 0.64G FLOPs) is over 4% better than that of MobileNet-BSEConv-based models, e.g., V1-BSEConv-U’s [29] obtained 59.1% with 3.33M, and 0.44G FLOPs. Meanwhile, about 2M parameters, TickNet-small (63.5%) is also 3.5% higher than V3-BSEConv-S’s. Please refer to Table 6 for more comparisons with other models.

4.4.2. Classification on ImageNet-1k and its challenging subsets

Due to the modest performance of the basic tick-shape network (see Fig. 11), we just deployed TickNet-small and TickNet-large on the enormous datasets (i.e., ImageNet-1k [46] and Places365

Table 7: Top-1 performance (%) on ImageNet-1k and Places365.

Architecture	Places365		ImageNet-1k	
	Top-1	#Paras	Top-1	#Paras
MobileNetV1 [18]	53.50	3.58M	70.80	4.23M
MobileNetV2 [19]	52.19	2.69M	69.70	3.51M
MobileNetV3 [20]	53.53	4.67M	71.50	5.48M
V1-BSConv-U [29]	-	-	71.50	4.23M
V2-BSConv-S [29]	-	-	69.80	3.51M
V3-BSConv-S [29]	-	-	71.50	3.93M
V1-GDF [44]	-	-	67.55	4.30M
V2-GDF [44]	-	-	67.12	3.58M
MobileNeXt [30]	51.48	2.72M	67.16	3.54M
ShuffleNetV2 [22]	50.80	1.63M	69.36	2.28M
ResNet18 [25]	54.43	11.36M	70.40	11.69M
ResNet18 [25] + SE [42]	-	-	70.59	11.78M
DenseNet-121 [62]	-	-	74.43	7.98M
EfficientNet-B0 [23]	-	-	77.10	5.29M
MnasNet [24]	-	-	75.20	4.38M
SqueezeNet [61]	-	-	41.90	1.25M
TickNet-small	54.34	2.35M	70.04	3.00M
TickNet-large	55.00	3.27M	72.08	3.92M

Note: “-” means “not available” or “not to be reported”, FLOPs and #Paras are computed by using tool *ptflops* [53].

[52]) for the following evaluations. It can be seen from Table 5 that the proposed networks obtained the highest rates on the challenging subsets of ImageNet-1k. Indeed, classifying the images of ReINs, TickNets are from 2% to 3% better than MobileNet-based networks [18, 19, 20, 30] and other light-weight models [22, 25]. Particularly, TickNet-small and TickNet-large are up to ~7% better than MobileNets on ImageNet-100. This advance is thanks to the efficient concentration of TickNets on representing the crucial attributes of objects in an image. [Fig. 12 shows an intuitive view of learning an image of geese, where TickNet-large has better concentration on the spatial information of geese in comparison with other light-weight networks.](#) In respect of image classification on ImageNet-1k, TickNets also gained good performance, 70.04% (3M) for TickNet-small and 72.08% (3.92M) for TickNet-large. These rates are competitive versus other small networks (see Table 7). It should be noted that DenseNet-121 [62], EfficientNet-B0 [23], and MnasNet [24] obtained better results. Nevertheless, they need larger #parameters on ImageNet-1k, while their performance is lower than TickNets’ on the other datasets (see Tables 6 and 5).

4.4.3. Classification on Places365 and its challenging subsets

Table 5 indicates the valuable performance of TickNets in comparison with other light-weight networks. For example, TickNet-large obtained 52.76% on RePL³⁰ and 53.43% on RePL⁵⁰, a little higher than the MobileNet-based models, e.g., MobileNetV2 [19] with 52.67% and 52.02%;

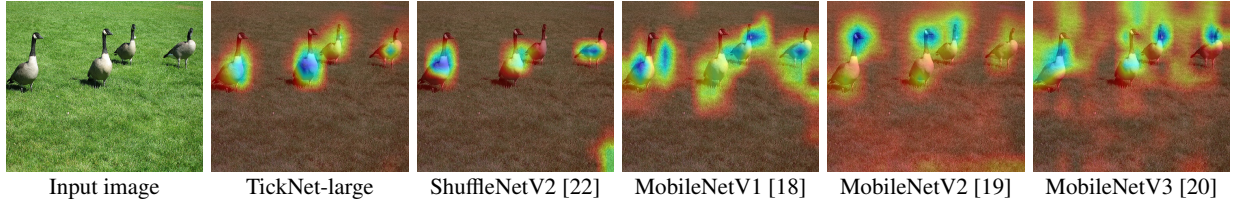


Figure 12: Efficiency of TickNet-large in learning intensify on the spatial features of geese in comparison with other light-weight networks. These visualizations are formed by using Grad-CAM [54].

MobileNeXt [30] with 51.24% and 52.17% respectively. Moreover, TickNet-basic also got comparative results with only 1M parameters. With respect to image classification on the large dataset (i.e., Places365 [52]), TickNets have good performance with 55% for TickNet-large, the highest rate versus the reported ones (see Table 7).

5. Conclusion

An efficient FR-PDP block has been presented in this work to take advantage of point-depth-point features extracted by a dynamic association of depthwise and pointwise convolutions with a full-residual mechanism. We then introduced a light-weight backbone based on a novel concept of channel elasticity of FR-PDP blocks as in a tick-shape. By hooking together tick-shape backbones, three light-weight TickNets were constructed correspondingly and then investigated the learning ability for image classification. The experimental results on benchmark datasets have proved the efficacy of FR-PDPs as well as the good performance of TickNets in comparison with other networks. For perspectives, other robust feature attention modules (e.g., BAM [47], CBAM [48], etc.) as well as dilated convolutions and spatial pyramid pooling techniques can be addressed for FR-PDP to investigate further improvement in practice. Such adjusted FR-PDP block can be considered for the deeper structure of large networks, e.g., ResNets [25], VGGNets [63], InceptionNets [64, 28], etc.

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